



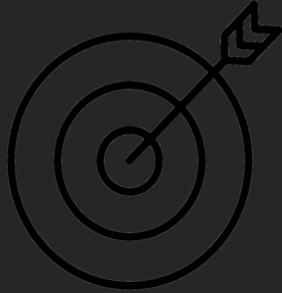
Educational Attainment In Maryland

ASMA ABBAS

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DATA 205: 32036

Overview



Goal

The main objective of this project is to discover the relationship between socioeconomic factors and educational attainment within Maryland counties.



Motivation

Education should be a right, not a privilege. Understanding the factors that impact educational attainment, as well as the demographics these issues are concentrated in, can bring us closer to solutions.



We'll be looking at...

An exploratory data analysis that covers general statistics based on educational attainment, correlation analysis between counties, predictive regression models, and more. This is a socioeconomic and educational inequality analysis.

Data

- Dataset 1

sourced from MarylandOpen Data, this dataset has information regarding different socioeconomic factors across Maryland counties. Some key variables include income, unemployment, and education levels.

- Dataset 2

Sourced from Census.gov, this dataset contains socioeconomic information for Montgomery County. Key variables include poverty rate, child poverty rate, median household income, and education levels.

- Dataset 3

Sourced straight from MCPS , this dataset contains specific information on high schools with Montgomery County. Key variables include, dropout rate, graduation rates, attendance rate, and FARMS participation.

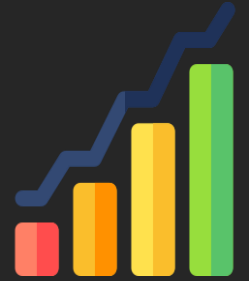


Tools Used and Data Cleaning



- Primarily, I used R studio for all my graphs and coding. I used ggplot and tidyverse, along with other packages like janitor for cleaning and graphs. I also made use of packages like tigris and sf for my maps.
 - To begin, I did some cleaning for the datasets:
 - Renaming variables and making them easier to work with, with the rename function.
 - Removing any missing values using is.na and na_omit.
 - Some of the variables were mislabeled as not numeric, so I had to work on fixing that as well, using as.numeric.
 - I created data frames specifically for graphs, and ran correlation analyses
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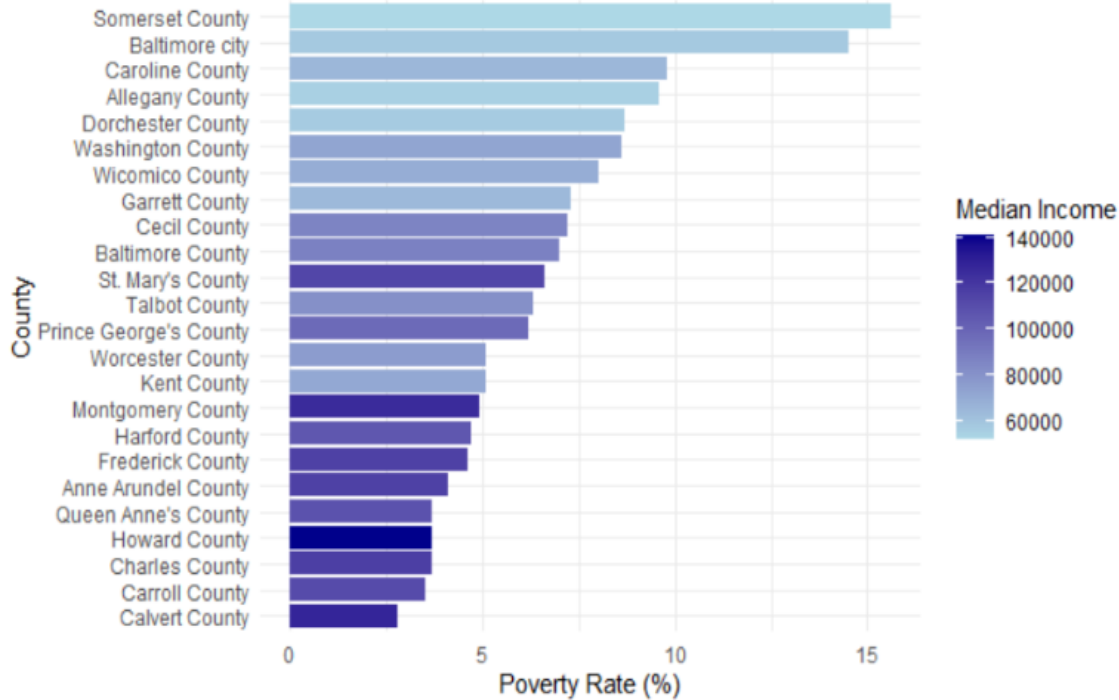
Summary Statistics



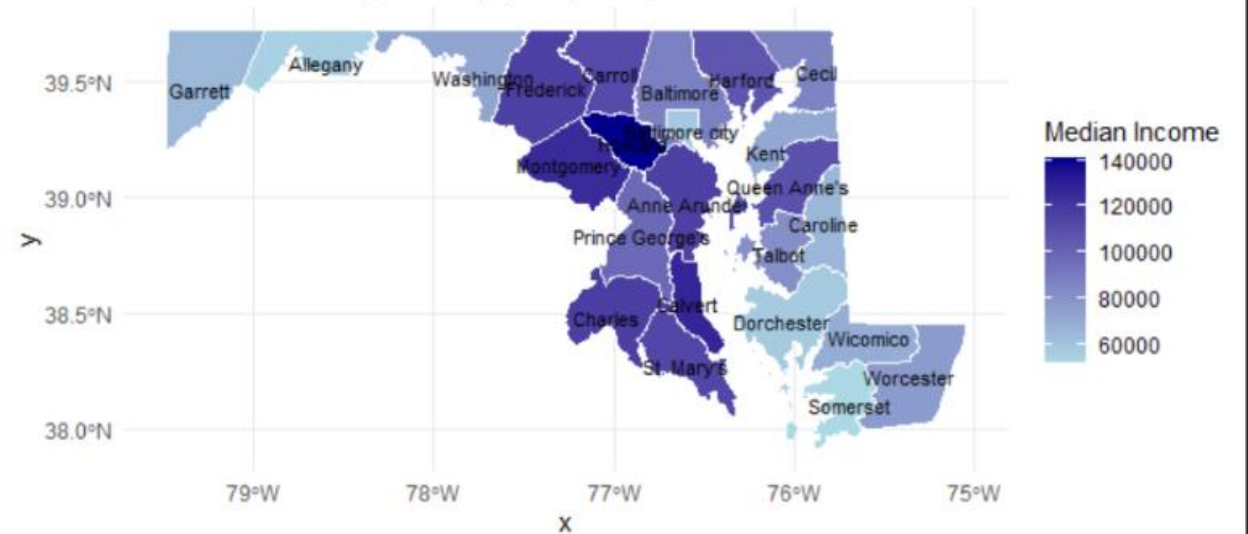
- Across counties:
 - Median County Income: 87,513
 - Mean county income: 90,906
 - Range: \$52,149 to 140,972
 - These rates make it clear that Maryland contains major income inequality across counties
 - Average poverty rate: 6.7%
 - Highest county: 15.6%
 - Lowest county: 2.8%
 - Average unemployment: 4.9%
 - Within MCPS:
 - FARMS:
 - Minimum: 6.4%
 - Mean: 39.17
 - Max: 65.9
 - SD: 1.848
 - There's variation in FARMS participation.
 - Average graduationrate:87.91
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EDA Visualizations

Poverty Rate by County (Colored by Income)

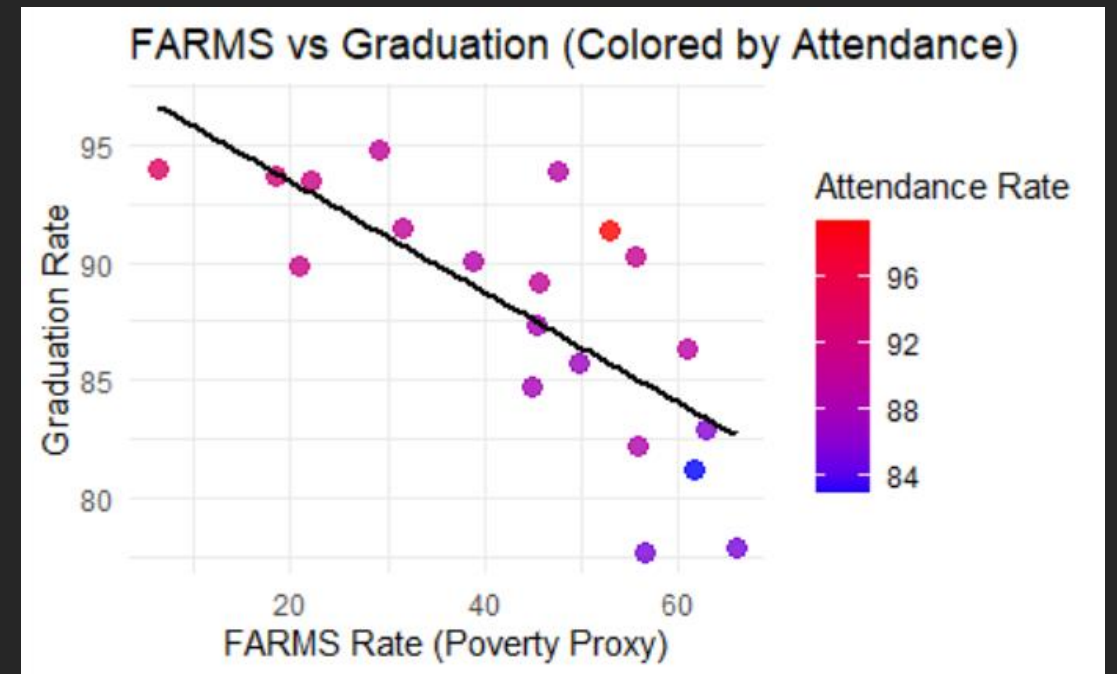


Median Income by County (Maryland)



Correlation Analysis

- What is the relationship between poverty and graduation rates?
 - Correlation = **-0.775**
 - There is a strong relationship, indicating higher poverty is linked to lower graduation rates
 - As poverty increases, graduation rates tend to decrease. The result is a strong negative.

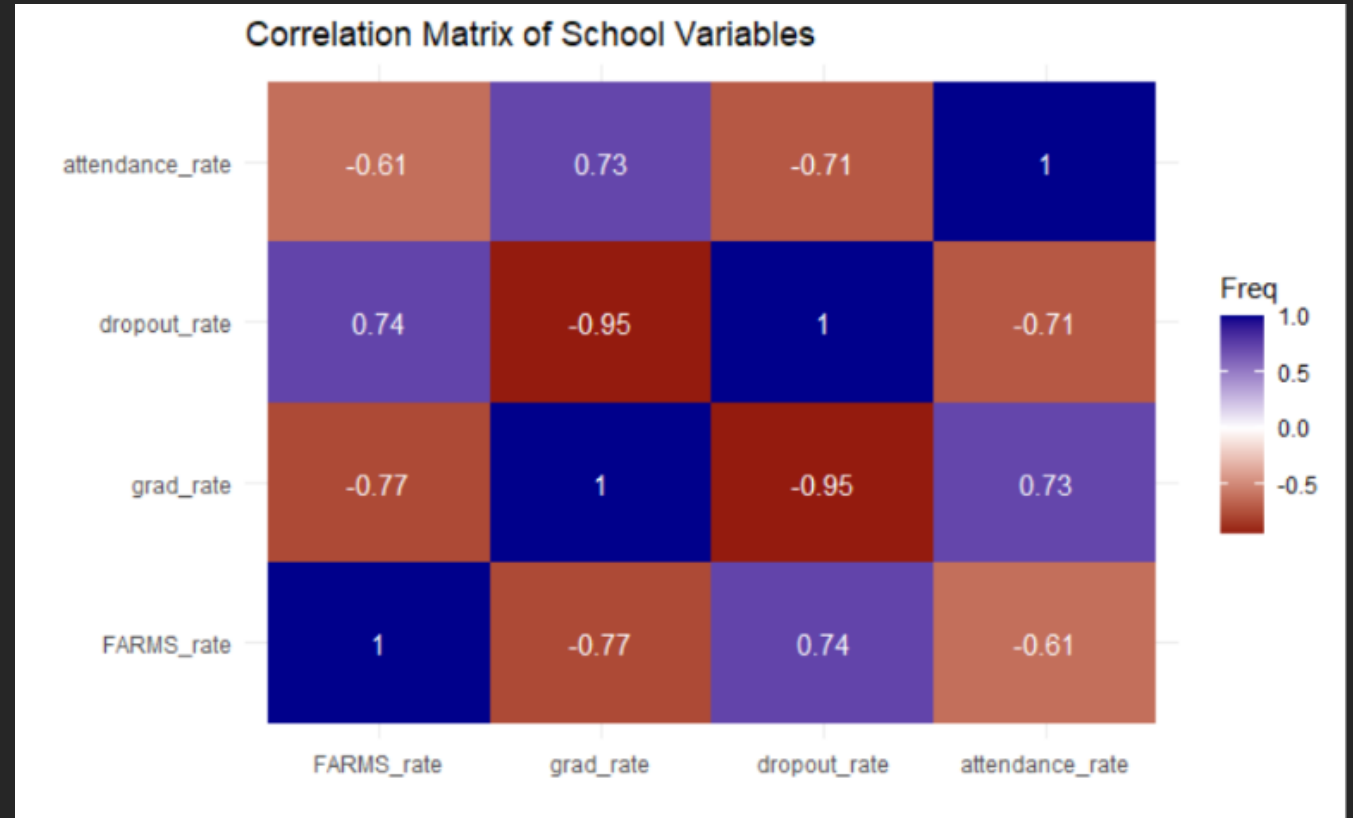


Correlation Analysis: other revelations

- Poverty vs Median income
 - **-0.791**
 - A strong negative relationship between the variables, representing how wealthier areas hold lower poverty rates.
 - Poverty vs unemployment
 - **0.659**
 - This number represents a generally strong positive relationship,
 - This shows that counties with a higher unemployment rate also hold higher poverty rates
 - **Income vs. Unemployment**
 - **-0.593**
 - Representing a generally negative relationship
 - Wealthier counties have lower unemployment levels
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Correlation Heatmap

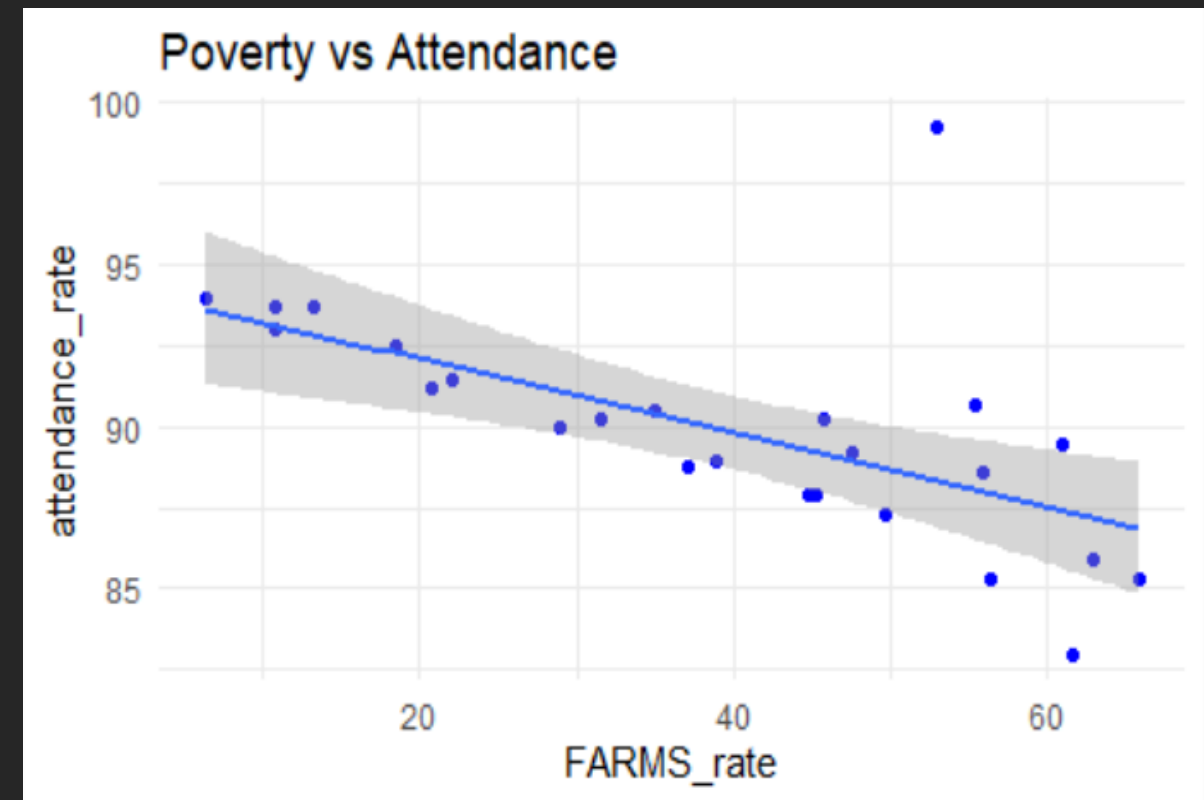
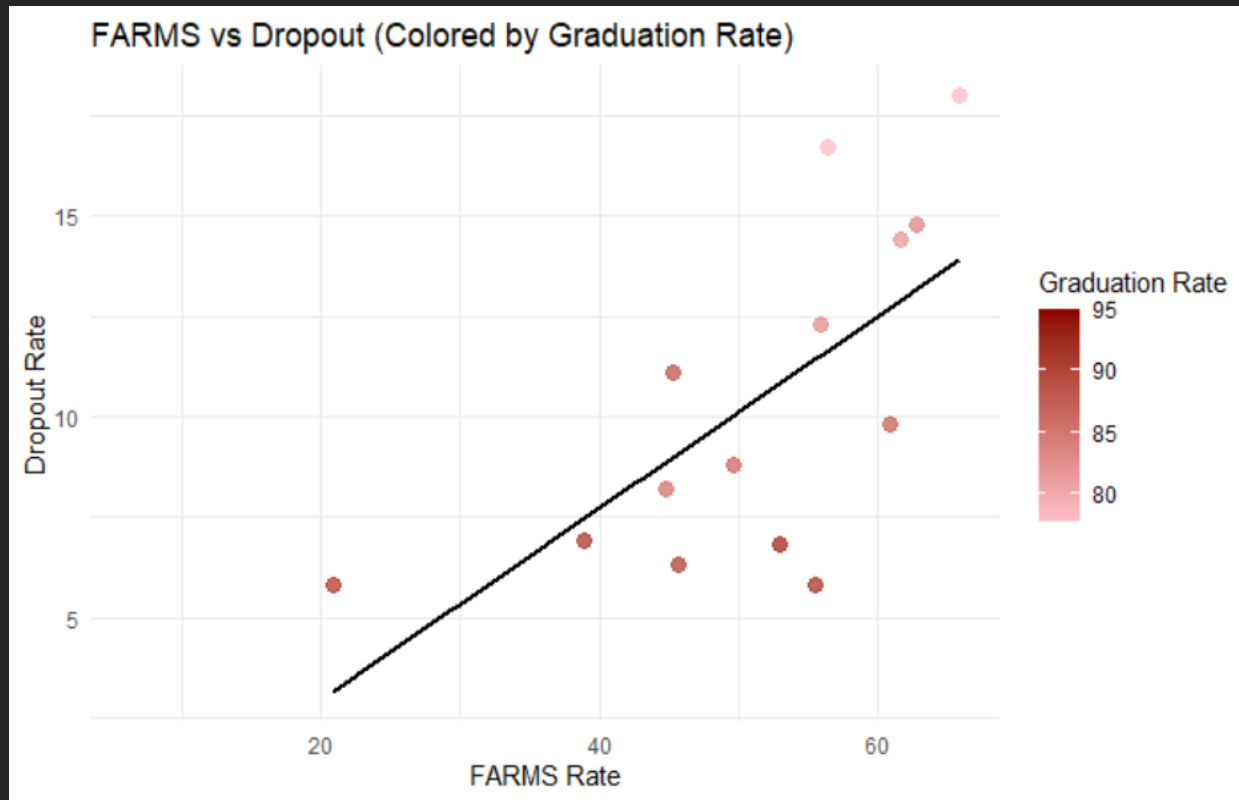
- Farms and dropout rates have a strong positive relationship with 0.74
- Farms rate and attendance have a moderately negative relationship, at -0.61
- Grad rate and dropout rate have an extremely strong negative
- Graduation rate and attendance have a strong positive at 0.73
- Dropout rate and attendance rate have a strong negative at -0.71





Key Insights

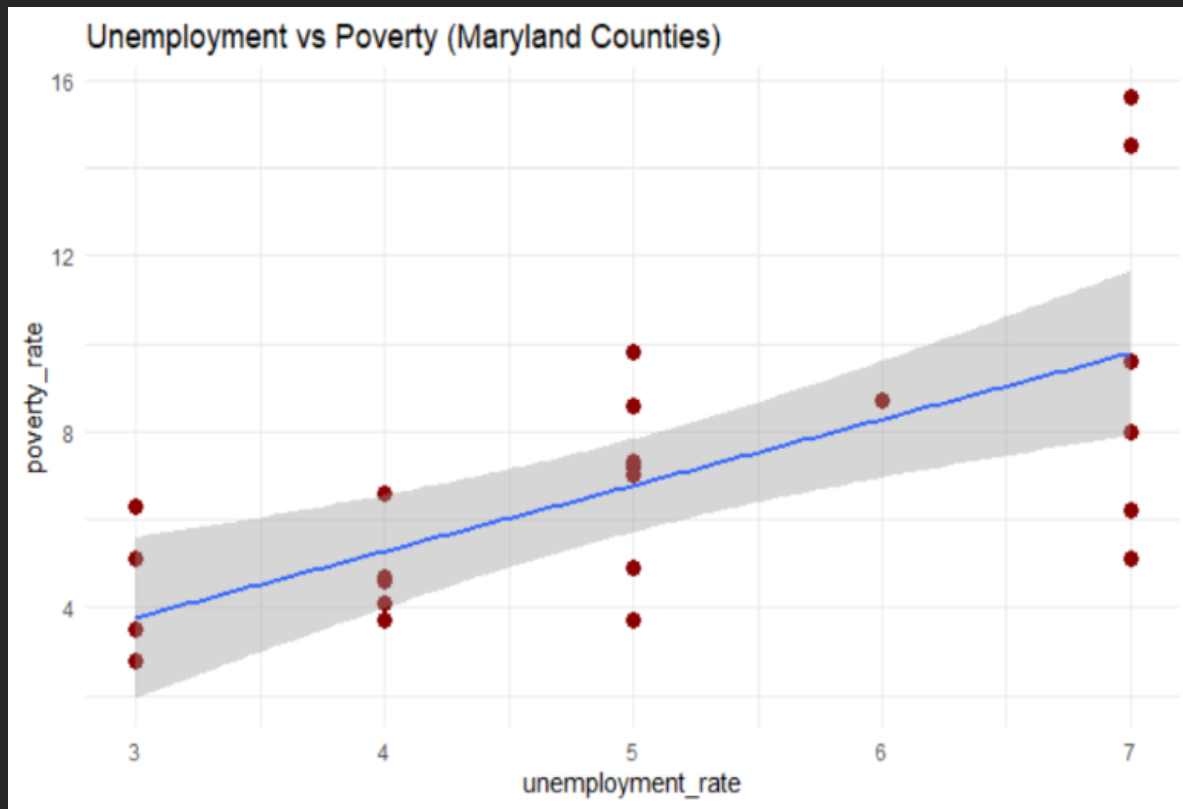
Impact of poverty on dropout rates, graduation rates, and attendance.



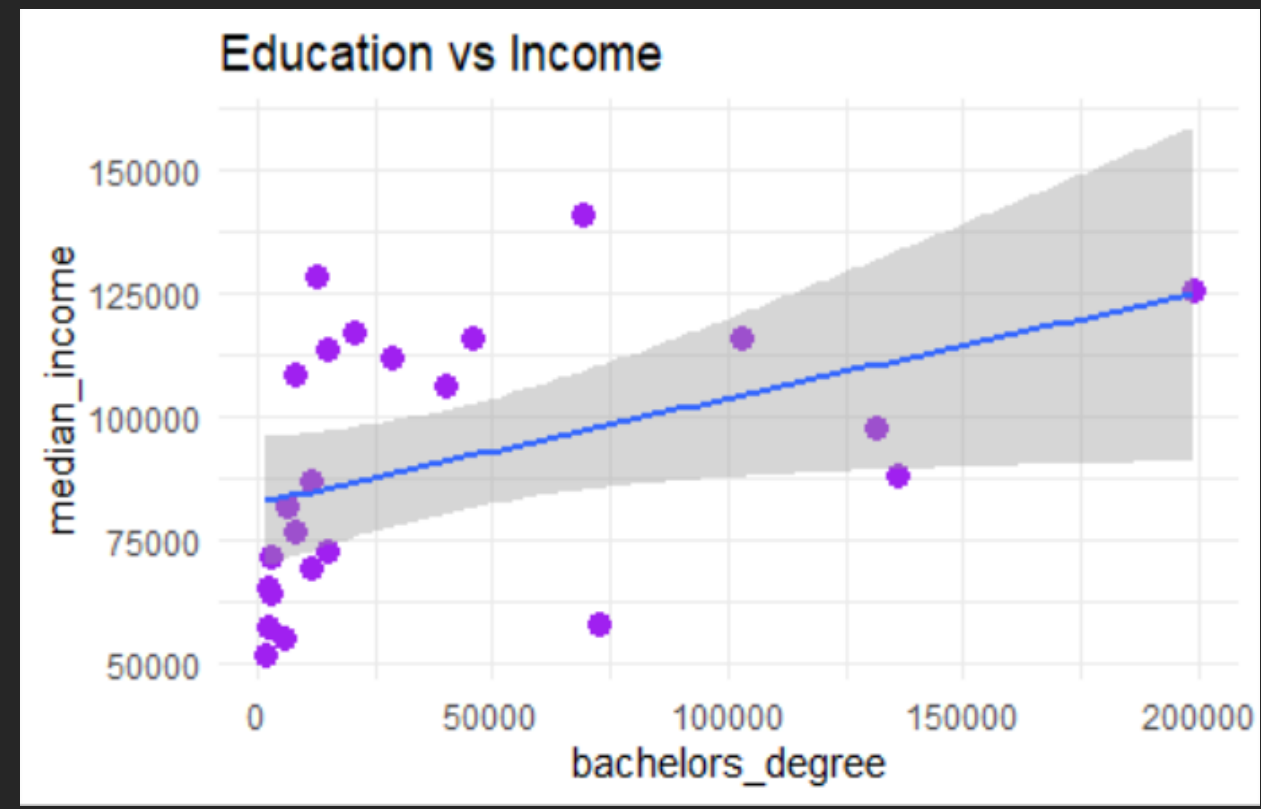
Key Insights



Unemployment vs Poverty

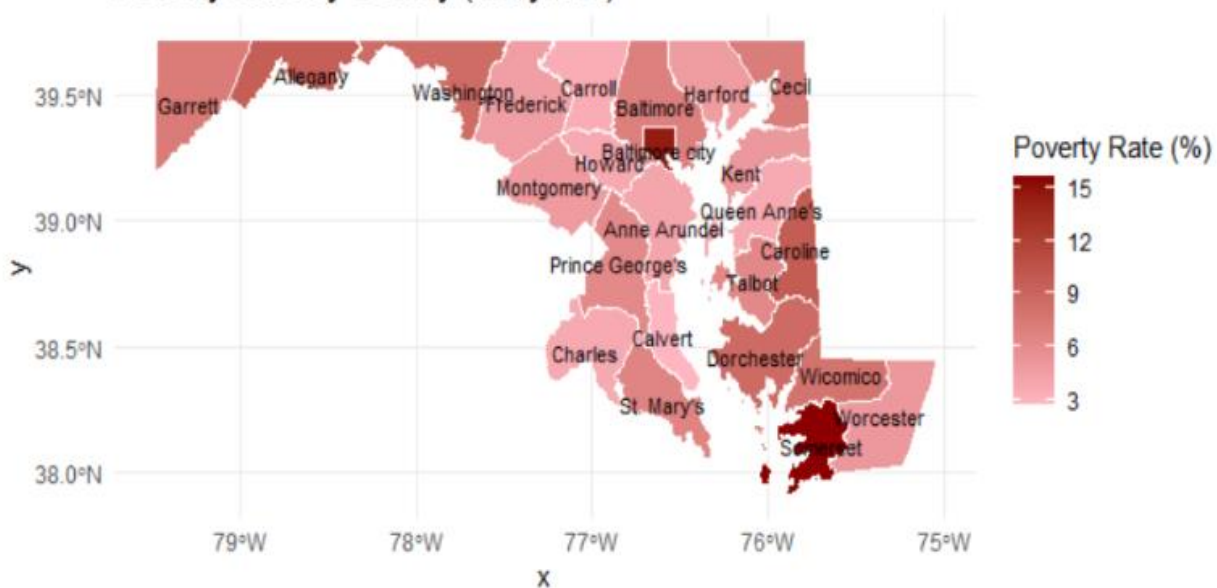


Education vs Income

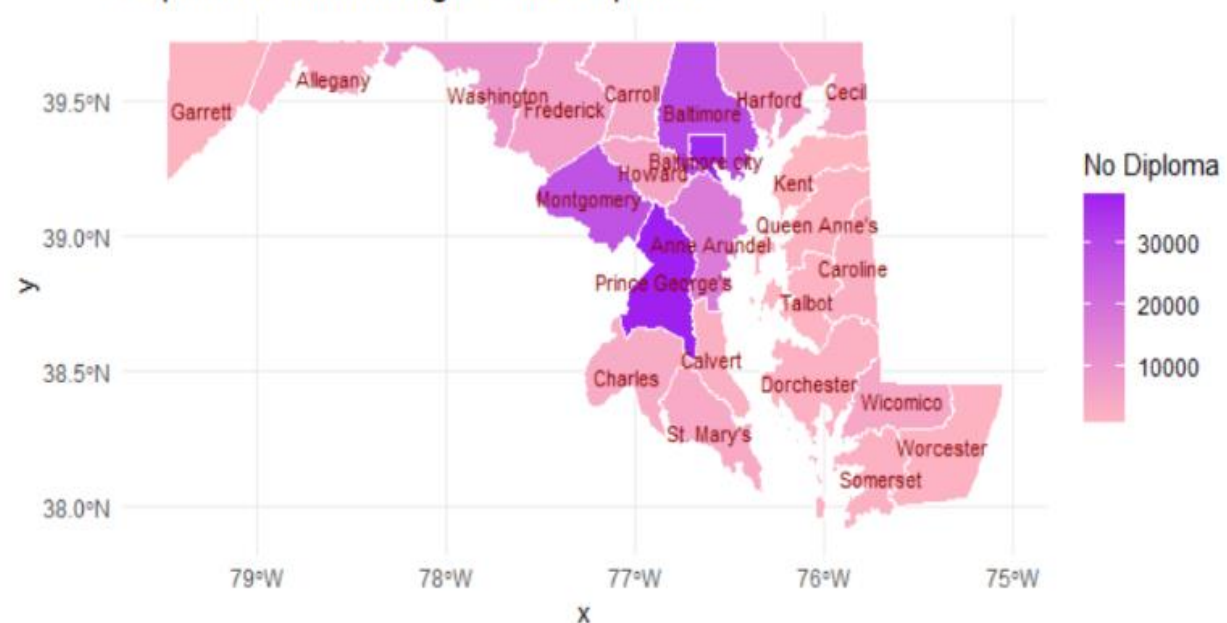


Heatmaps

Poverty Rate by County (Maryland)



Population Without High School Diploma

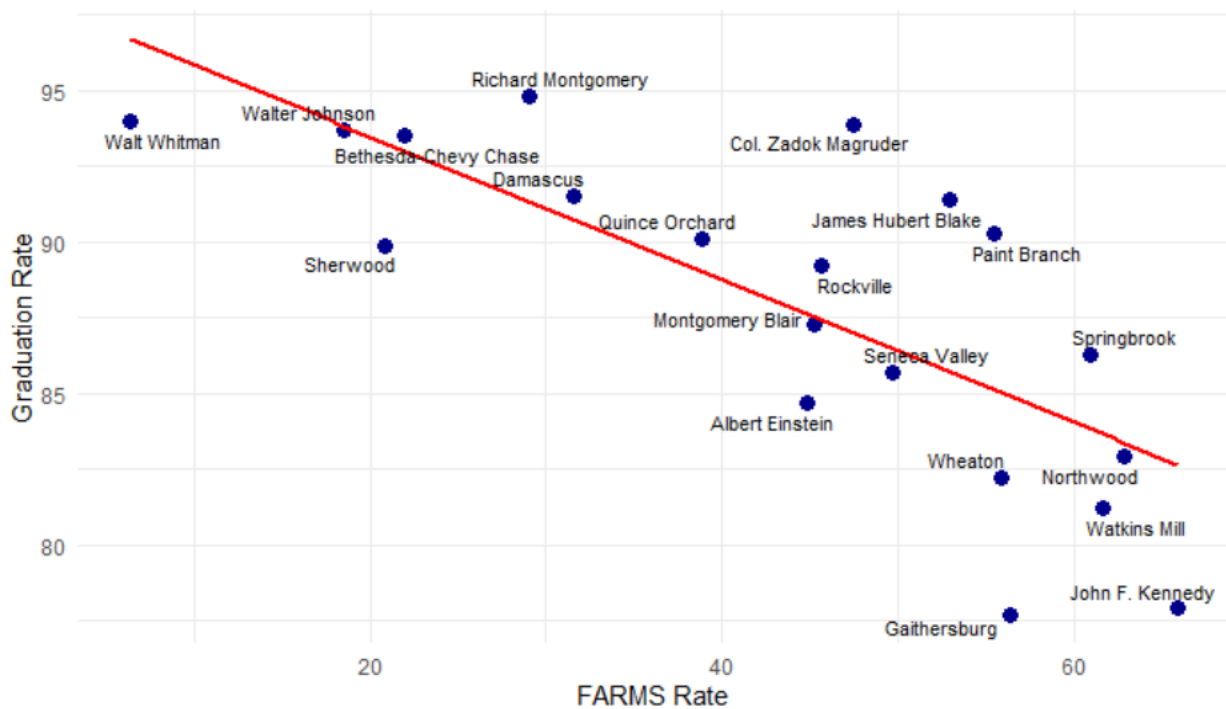


EDA: visualizations

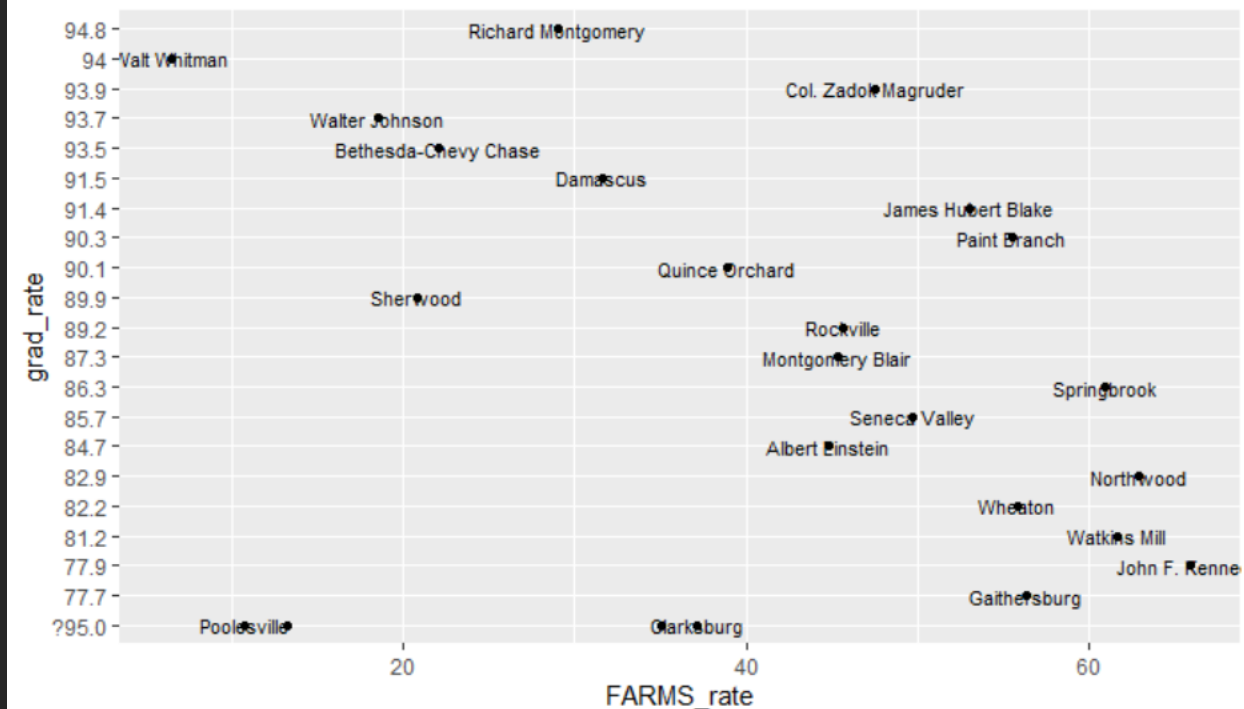
Performance across schools



Schools Performing Above or Below Expectations

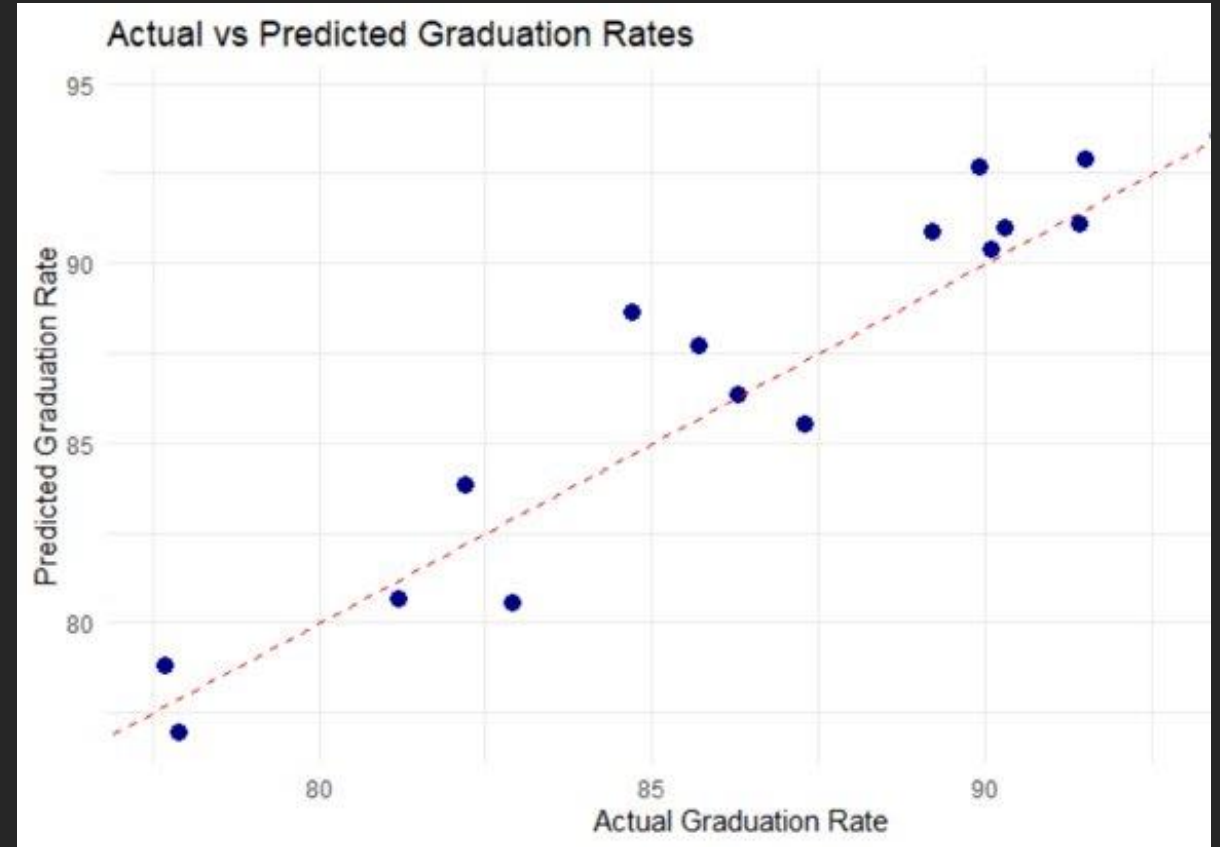


Outlier Schools



Model

- For this model, I wanted to predict graduation rates. I proceeded with this by using factors like:
 - poverty indicators, attendance, and dropout rates.
 - The result of the r-squared test was 0.912, meaning that the model explains 91% of graduation rate variation.
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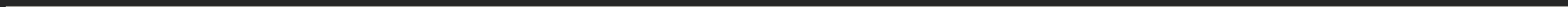


Conclusion



- ❖ The findings from this analysis indicate educational inequality is closely tied to broader socioeconomic conditions. Poverty has its role to play in lower rates of educational attainment.
 - ❖ **So what can we do about this?**
 - Focus on areas that are most disadvantaged, learning from areas that appear balanced with their poverty and educational outcomes.
 - Argue for policy change and accessible education, especially for kids coming from economically disadvantaged homes
 - Create other support, similar to FARMS but in more academic departments.
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Thank You!



Acknowledgements and References

- Thank you to Professor Perine for a wonderful and informative semester, and my classmates for constructive feedback. I was able to learn so much in this class, and by watching my classmates.
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